Lab Assignment 8

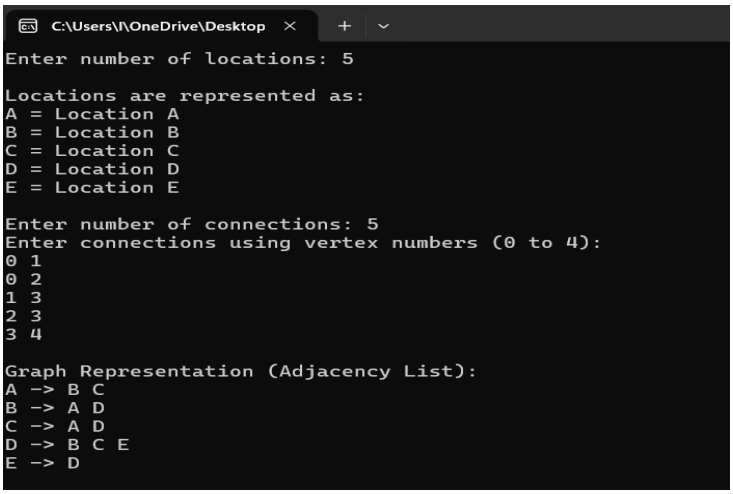
Title : Consider a particular area in your city. Note the popular locations A, B, C .

. . in that area.

Assume these locations represent nodes of a graph.

Program :

```
#include<iostream>
#include <vector> using
namespace std;

int main() {
    int n, edges;

    cout << "Enter number of locations: "; cin >>
    n;

    vector<vector<int>> graph(n);

    cout << "\nLocations are represented as:\n"; for (int i
    = 0; i < n; i++) {
        cout << char('A' + i) << " = Location " << char('A' + i)
             << "\n";
    }

    cout << "\nEnter number of connections: "; cin >>
    edges;

    cout << "Enter connections using vertex numbers (0 to "
         << n - 1 << "):\n";

    for (int i = 0; i < edges; i++) { int u,
        v;
        cin >> u >> v;
```

```
        if (u >= 0 && u < n && v >= 0 && v < n && u != v) {
            graph[u].push_back(v);
            graph[v].push_back(u);
        } else {
            cout << "Invalid connection!\n";
        }
    }
    cout << "\nGraph Representation (Adjacency List):\n"; for (int i = 0; i < n; i++) {
        cout << char('A' + i) << " -> ";

        for (int j = 0; j < (int)graph[i].size(); j++) {
            cout << char('A' + graph[i][j]) << " ";
        }
        cout << "\n";
    }
    return 0;
}
```

Output: